ADI TFLITE MICRO SHARCFX

Product Description

ADI TFLITE Micro SHARC-FX is a library built using Google's TFLM (TensorFlow Lite for Microcontrollers) front-end with custom optimizations for select kernels enabling massive performance improvements as compared to the reference TFLM kernels. It has been tested on the EagleHP chips, ADSP 21835 and SC-835.

Product Highlights

ADI TFLITE Micro SHARC-FX module has been highly optimized to run on the EagleHP hardware using SHARC-FX intrinsics, to achieve performance improvements over vanilla TFLM.

To optimize the performance, bugs were fixed, and best practises were used according to industry standards.

This release supports optimization support for select kernels and features few example applications that the optimized library can enable running on SHARC-FX based chips (EagleHP).

The module provides a framework for users to get familiar with the library and encourages them to use it as part of their own applications, using the example applications as reference. The example applications contain projects that work on both file IO input as well as real-time input. This release features 5 applications (audio DTLN denoiser, keyword spotter, genre identification and Urbansound classification, audio DFN denoiser) that work both in File I/O and real-time modes.

Cycle count Summary Table

The cycle count results for the library kernels vary for different input and kernel configurations, a handful of which are summarized in the table below. The second table also describes the cycle counts for the applications that are real-time ready. The numbers mentioned are obtained from running the library/applications with the simulator in CCES. We have verified that the simulator numbers closely match the numbers we get from running them on the board(emulator).

Currently, there are 2 configurations of Convolution2D kernel, the kernel that gets selected based on the condition of which gives us the most speedup in each case.
Newer versions [starting with rel4.0.0] of the pointwise convolution and the fully connected kernel expect weights to be in a non-standard format to fully leverage the DSPs memory access pattern to achieve higher performance. To use the new optimizations in applications using these layers, the model weights need to be reordered offline to match the format of weights expected by the layers, the same can be done with a tool provided which transforms the weights to the format expected by the supported kernels. Refer to the ADI_TFLITE_MICRO_SHARCFX_UsersGuide.pdf for more information.

CONV2D Generic Conv2d							
InHeight	**InWidth**	**InChannels**	**FilterWidth**	**FilterHeight**	**FilterCh**	**FilterStride**	**CycleCount**
128	128	1	3	3	64	1	**13.723069 MCycles**
32	32	8	3	3	64	1	**1.584605 MCycles**
32	32	16	3	3	64	1	**3.073143 MCycles**

CONV2D With RADD							
InHeight	**InWidth**	**InChannels**	**FilterWidth**	**FilterHeight**	**FilterCh**	**FilterStride**	**CycleCount**
128	128	1	3	3	64	1	**309.191562 MCycles**
32	32	8	3	3	64	1	**18.992811 MCycles**
16	16	512	3	3	64	1	**11.907781 MCycles**

DEPTHCONV2D							
InHeight	**InWidth**	**InChannels**	**FilterH**	**FilterW**	**FilterCh**	**FilterStride**	**Cycle Count**
25	5	64	3	3	64	1	**0.067231 MCycles**
13	5	276	3	3	276	1	**0.147619MCycles**

Fully Connected

InNeurons	OutNeurons	Cycle Count
2048	1024	**0.852112 MCycles**
256	10	**0.152445 MCycles**
64	12	**0.002094 MCycles**

LSTM				
Batch	**Timesteps**	**Input features**	**State dimension**	**Cycle Count**
2	3	2	2	**0.018977 MCycles**
1	5	20	10	**0.028556 MCycles**
1	10	20	20	**0.09395 MCycles**
1	5	50	10	**0.030291 MCycles**

Application	Cycle Count
Keyword Spotter (per forward pass*)	**1.359 MCycles**
DTLN Denoiser (per forward pass*)	**1.029 MCycles**
Genre Identification (per forward pass*)	**2.6 GCycles**
Urbansound Classification (per forward pass*)	**2.6 GCycles**
DFN[DeepFilterNet] Denoiser (per forward pass*)	**1.43 MCycles**

(*) Keyword Spotter – 1 forward pass takes in 1 s of audio as input at 16KHz sampling rate.

(*) DTLN Denoiser – 1 forward pass takes in 32 ms of audio as input at 16KHz sampling rate.

(*) Genre Identification – 1 forward pass takes in 2.7 s of audio as input at 48000 Hz sampling rate.

(*) Urbansound classification – 1 forward pass takes in 2.7 s of audio as input at 48000 Hz sampling rate.

(*) DFN denoiser – 1 forward pass takes in 8ms of audio as input at 48000 Hz sampling rate.

Deliverables

- ADI TFLITE Micro SHARC-FX Release package contains the optimized NN library with the TFLM frontend. It also contains the example applications both with File IO and real-time input for DTLN Denoiser, Keyword Spotter, Genre Identification, Urbansound Classification and DFN denoiser.
- Detailed documentation, for the user to get familiar with the project in the Getting Started Guide.
- Detailed steps to run the example applications on the simulator (in CCES) and emulator (board) and visualize the results in the User Guide.

Availability

This module is available in source code format after acceptance of ADI's Software License Agreement.

For support on any of our software modules, please contact local FAE.

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DFN Licensing Disclaimer:

The Deep Filtering Network (DFN) model is subject to licensing restrictions and cannot be used in commercial products without prior approval. If you require a commercially licensed version of the DFN model, please contact the ADI Eagle-NN team to request a commercially approved model before use.